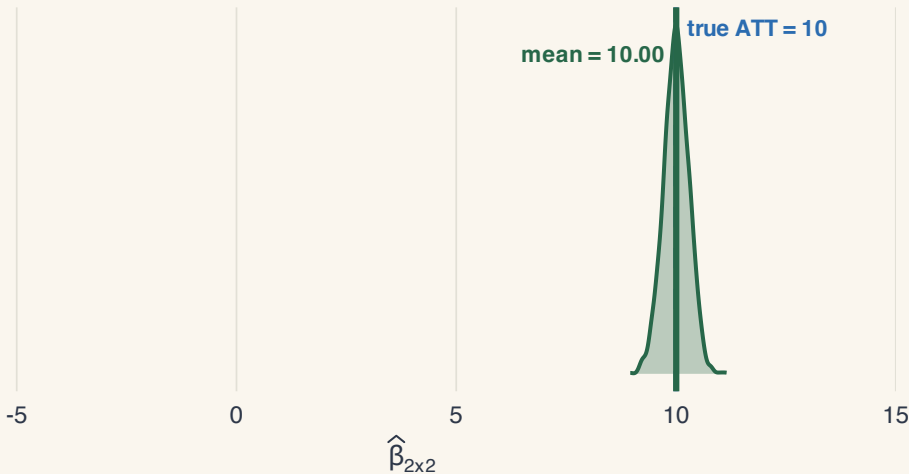


Four ways the 2x2 can go wrong

Sampling distribution of $\hat{\beta}$ over 2000 Monte Carlo reps · n = 200 per group · True ATT = 10

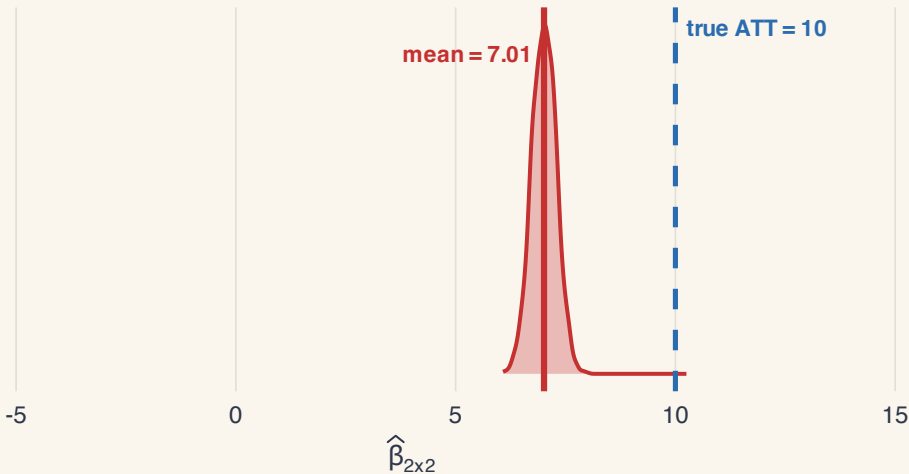
Case 1. NA holds, never-treated control

Predicted: beta = ATT = 10



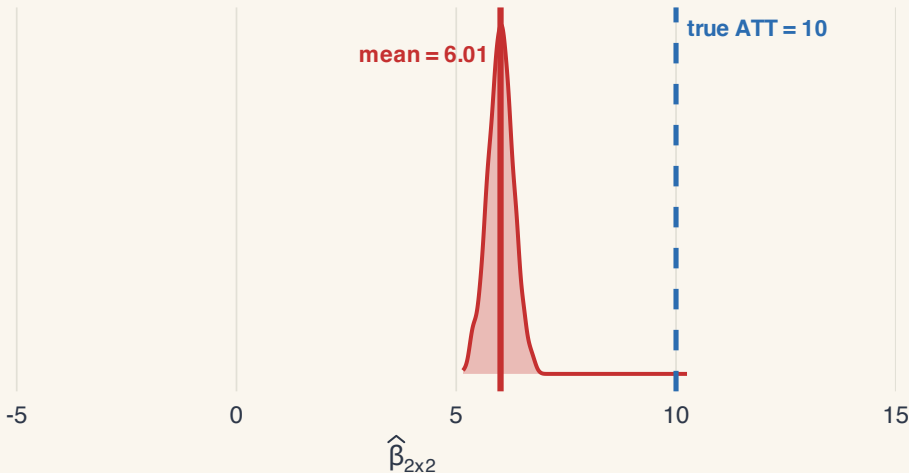
Case 2. NA fails (anticipation = 3)

Predicted: beta = ATT - ATT_pre = 10 - 3 = 7



Case 3. Spillover into control (= 4)

Predicted: beta = ATT - spillover = 10 - 4 = 6



Case 4. Already-treated control (Δ ATT = 14)

Predicted: beta = ATT - Δ ATT = 10 - 14 = -4

